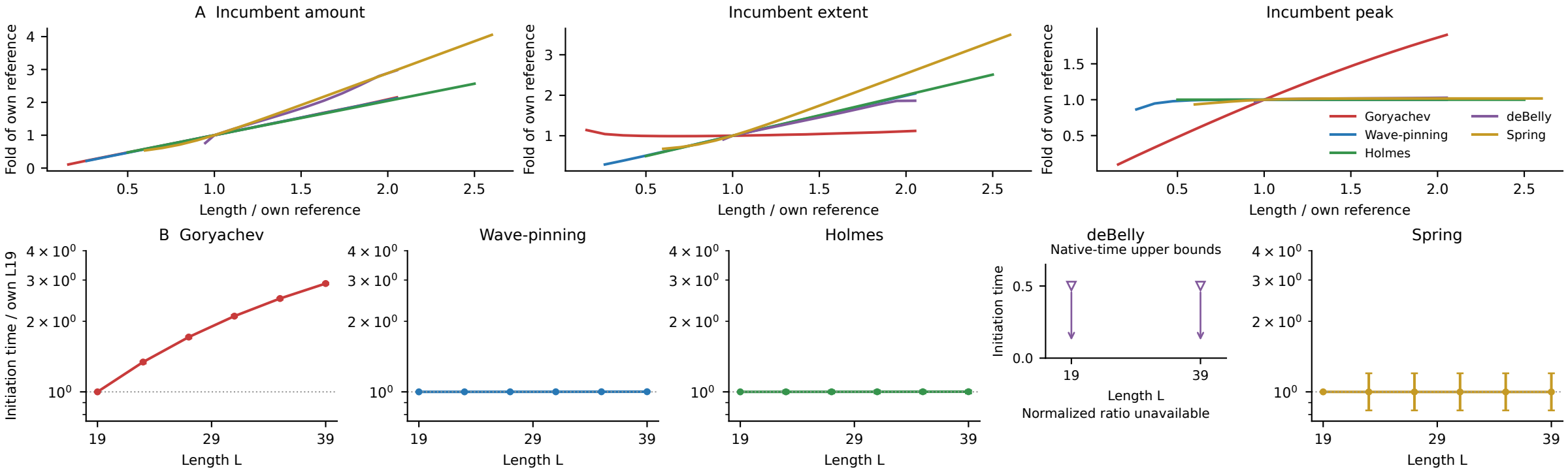
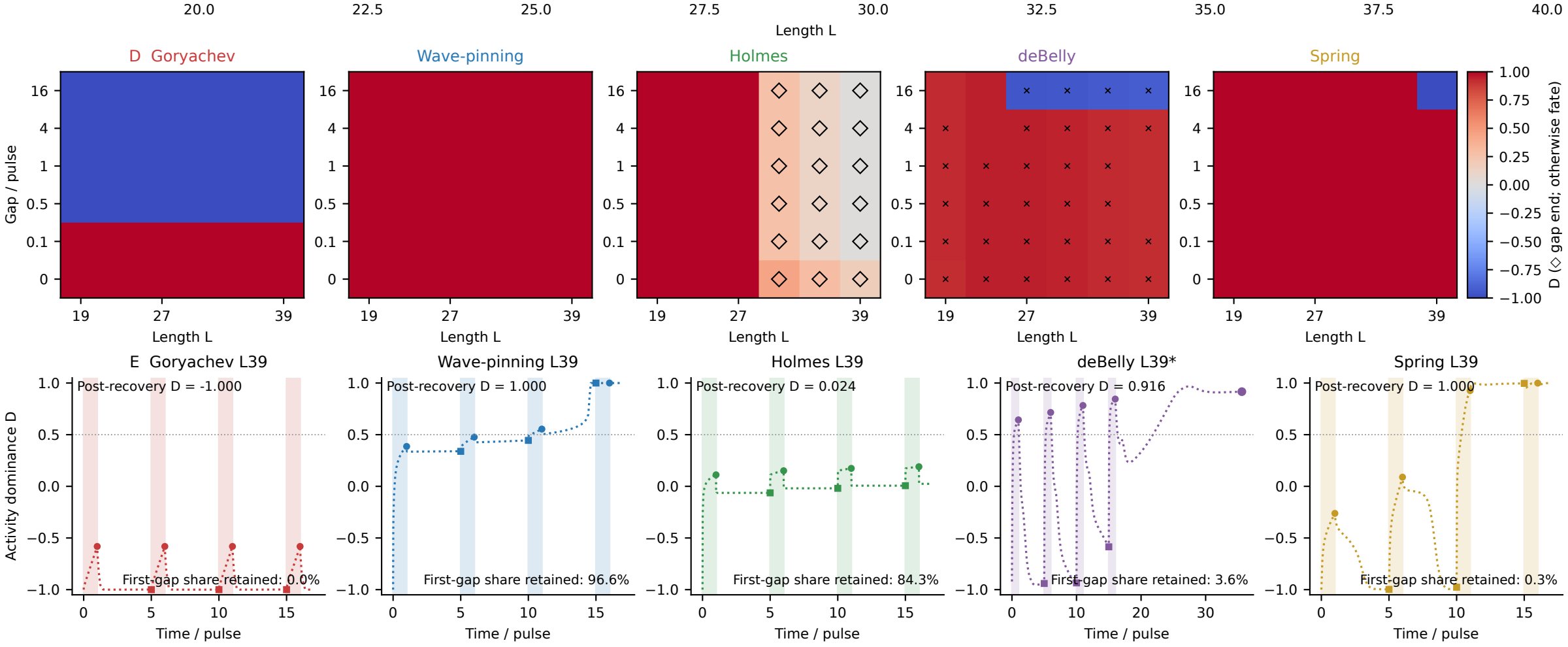
# Replacement resistance in enlarged cells arises through three mechanism classes; cue shape and timing distinguish their signatures




F Shape and timing signatures distinguish mechanism classes

| Class / models | Initiation slows | Equal-area penalty | Share through gaps | Commit after collapse | Experimental measurement |
|---|---|---|---|---|---|
| Initiation-limited Goryachev | Yes (2.90×) | Modest (1.12×) | Resets | No | Beam width at fixed photon dose; pulse trains; gap-end biosensor share |
| Mass-conserved, saturating Wave-pinning, Holmes | No resolved slowing | Large; grows over resolved L range | Share retained (ratchet) | — | Beam width at fixed photon dose; pulse trains; gap-end biosensor share |
| Slow non-activity state deBelly*, Spring | Unresolved* / no | Modest (~1×) | Collapses early | Yes | Beam width at fixed photon dose; pulse trains; gap-end biosensor share |

◇ Holmes L31–L39: last inter-pulse gap-end D, with no target-length calibration. × failed field tail; ? uninformative ratio.

* deBelly initiation is interval-censored; later-gap timing and endpoint fields carry numerical qualifications (complete supplement).